AI Personal Study Assistant

Project Documentation

Student Name	Charles Josemon
Department	Computer Science
College	TKM IT
Subject	Artificial Intelligence and Personalized Learning
Website Link	https://idea-to-inkwell-studio.lovable.app/

1. Introduction

An **AI Personal Study Assistant** is an AI-powered tool, usually a chatbot, that helps students learn by answering questions, explaining topics in simple language, summarizing notes and creating practice questions. This project builds a working AI Personal Study Assistant using study material in PDF format, a chatbot platform and a website. The assistant answers student questions using the uploaded study material as its knowledge base.

2. Objectives

- To create clear, well-organized study material in PDF format that can serve as a chatbot knowledge base.
- To build an AI chatbot that answers student questions from that study material.
- To create an interactive website where students can use the chatbot and other study tools.
- To publish the website and make the assistant available to students through a link.

3. Project Steps and Work Done

Step 1: Creating the Study Material PDF

A study material document was created in PDF format. It has clear headings, defined technical terms, comparison tables, direct answers to common questions, and review questions with answers, so the chatbot can find and use the information easily.

PDF	Content
AI Personal Study Assistant Study Material	What AI and AI study assistants are, key features, technologies (AI, machine learning, natural language processing, large language models, retrieval-augmented generation, PDF processing, chatbot technologies), system architecture and workflow, tools and platforms, advantages, limitations and applications, conclusion, review questions and FAQs.

Step 2: Creating the Chatbot

The PDF was uploaded to **Chatbase**, a platform for building custom AI chatbots that answer questions from your own data. Chatbase processed the file into a knowledge base. When a student asks a question, the system finds the parts of the PDF closest in meaning to the question and the language model uses them to write the answer. This approach is called **Retrieval-Augmented Generation (RAG)**.

Step 3: Creating the Website

A prompt describing the required website was written and given to an AI website builder (Lovable). The builder generated an interactive study dashboard titled "AI Personal Study Assistant". The generated website was checked to make sure all sections displayed properly.

Section	Purpose
Overview	Introduces the assistant and its main features: smart notes, adaptive quizzes, instant answers and study planner.

Section	Purpose
Flashcards	Flip cards for quick revision of important questions and answers.
Mini Quiz	Multiple-choice questions with instant feedback and a score counter.
Study Timer	A Pomodoro timer that helps students study in focused sessions with short breaks.
AI Chat	The section where the Chatbase chatbot is placed so students can ask questions.
Dark Mode	A toggle that changes the website theme for comfortable viewing.

Step 4: Deploying the Chatbot and Publishing

The Chatbase chatbot was added to the website using Chatbase's supported embed option (embed code or integration), so students can chat with the assistant inside the AI Chat section. The website was then previewed, tested and published, and the link was copied.

Published website: https://idea-to-inkwell-studio.lovable.app/

Screenshot of the published website: [Insert screenshot here]

Screenshot of the chatbot answering a question: [Insert screenshot here]

4. Tools and Technologies Used

Tool / Technology	Purpose in the Project
ChatGPT / AI assistants	Helped draft and refine the study material and website prompts. Content was reviewed for accuracy.
Microsoft Word / Canva / PDF	Formatting the study material and exporting it as a PDF used as the knowledge source.
Chatbase	Building the chatbot and knowledge base from the uploaded PDF.

Tool / Technology	Purpose in the Project
Lovable (website builder)	Generating the interactive website from a written prompt.
Large Language Model (LLM)	Understands questions and generates answers.
Retrieval-Augmented Generation (RAG)	Retrieves relevant parts of the PDF before the answer is generated.
Natural Language Processing (NLP)	Lets students ask questions in normal sentences.

5. System Workflow

1. The study material is prepared and exported as a PDF file.
2. The PDF is uploaded to Chatbase, which extracts the text, splits it into chunks and builds the knowledge base.
3. The chatbot is embedded in the website.
4. A student opens the website and types a question in the AI Chat section.
5. The system finds the most relevant chunks from the knowledge base.
6. The language model uses those chunks to write a clear answer.
7. The student reads the answer and can ask follow-up questions.

6. Sample Questions to Test the Chatbot

- What is an AI Personal Study Assistant?
- What is Retrieval-Augmented Generation (RAG)?
- What are the limitations of AI study assistants?

7. Advantages and Limitations

Advantages

- Gives instant answers at any time, day or night.
- Explains topics in simple language and supports personalized learning.
- Saves time by finding information quickly in large study material.
- Combines chat, flashcards, quizzes and a study timer in one place.

Limitations

- AI-generated answers can sometimes be inaccurate, so important facts should be checked with textbooks or teachers.
- The chatbot can only answer well about topics included in its knowledge base.
- Students should avoid sharing sensitive personal information in the chat.

8. Conclusion

This project successfully built an AI Personal Study Assistant from start to finish. Study material was prepared as a well-structured PDF on AI Personal Study Assistants. This PDF was used to build a chatbot in Chatbase that answers student questions from the uploaded content. An interactive website with flashcards, a quiz, a study timer and an AI chat section was then created and published, so students can access the assistant through a single link.

The project shows how modern AI technologies such as large language models, natural language processing and retrieval-augmented generation can be combined with simple tools to support education. It also showed the importance of good-quality, clearly written source material, because the chatbot is only as accurate as the documents it learns from. AI answers should still be verified, since AI can make mistakes.

In the future, the assistant can be improved by adding more subjects, supporting voice and multiple languages, tracking student progress, and showing the source of each answer. AI study assistants have strong potential to make learning more accessible, personalized and available to every student.

9. References

- Chatbase official website and help documentation: https://www.chatbase.co
- OpenAI documentation: https://platform.openai.com/docs
- Anthropic documentation: https://docs.claude.com
- Google AI for Developers: https://ai.google.dev
- Lovable website builder: https://lovable.dev